

LAYER 2 SPECIFICATION · SENSOR NOISE PROFILING

Layer 2 - Camera Sensor Noise Profiling (PRNU)

Owner: Arihant · Module: camera_sensor_noise_profiling.py

This note summarises the implementation. Rationale, measurements and limitations are in [ARCHITECTURE.md](#) (Sections 4.3, 6 and 7).

Model

A CMOS sensor outputs $I = I_0 (1 + K) + \theta$, where K is the pixel-wise PRNU gain pattern and θ is temporal noise. K is locked to the pixel grid and differs between sensors (Lukáš, Fridrich & Goljan 2006).

Pipeline

- Residual** $W = I - F(I)$. F is a spatially adaptive local Wiener filter: local variance is the minimum over 3/5/7/9 windows (Mihçak-style), and residual is attenuated by σ^2/var with $\sigma = 3$. Saturated pixels are zeroed.
- Fingerprint (MLE)**: $K = \Sigma W \cdot I / \Sigma I^2$, then row/column zero-meaning and DFT-domain peak clamping. Peak clamping removes the JPEG/H.264 block grid and CFA periodicities that every camera and codec share.
- Reference mode (enforced)**: correlate the clip's summed residual with $I \cdot K_{\text{ref}}$ and score the **PCE**, taken at the strongest peak within ± 2 px of zero shift (session-to-session crop offsets). The clip matches the enrolled camera if $\text{PCE} \geq 60$. Fingerprints are stored per **Layer 1 camera ID + sensor mode** (`FingerprintStore`), so the PRNU check is bound to the attested physical device.
- Blind mode (advisory)**: estimate K_A and K_B from temporally disjoint halves and correlate them **only on pixels whose low-passed content changed between the halves**. Significance comes from a circular-shift null (48 shifts) → z-score.
 - PRESENT: $z \geq 5$ and $p \geq 0.05$
 - ABSENT: $z < 3$ or $p < 0.04$
 - INCONCLUSIVE: less than 4% of the frame changed (highlights above 230 excluded), or anything in between

Why the previous version produced false positives / false negatives

- Odd/even frame interleaving plus whole-frame correlation measured **static scene texture**, not PRNU. Real webcams scored PCE 7k–63k, and the Grok AI video scored 79k and passed.
- The app saved live frames with OpenCV `mp4v` before analysis, and file mode re-encoded a "fast sample" the same way. That re-encode erases PRNU: genuine frames dropped from PCE 300+ to 14–22. All analysis now runs on raw decoded frames.
- A blind "is there a noise pattern?" test cannot be a hard gate. Native phone videos (stabilisation, ISP denoising) often show none, while generators and face-swap backgrounds often show one (see the benchmark). Only the reference match is enforced.

CLI

```
python camera_sensor_noise_profiling.py --video clip.mp4 # blind
python camera_sensor_noise_profiling.py --video calib.mp4 --enroll-ref fp.npy # enroll
python camera_sensor_noise_profiling.py --video clip.mp4 --ref-fingerprint fp.npy
```